

# CONNECTIVITY That Saves Lives



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**I**magine being at a hospital a few decades back. There is an emergency at the doorstep. Doctors and nurses are stepping in to attend to the situation. Counsellors are trying to calm people down associated with the affected patient. Constant real-time monitoring of the status was something that was manually done by experts. Randomness and confusion prevailed.

Enter today; the term real-time monitoring is very much possible, thanks to technology. It is the evolutionary advantage that gives us the edge to achieve more with the invisible fabric of connectivity that envelops us—the Internet. If you thought the Internet was an information bank, its brainchild—the Internet of Things (IoT) is another form of it, sweeping off information from things and helping us learn what we are and what we could do with data.

India has seven doctors for every 10,000 people, and data from Indian Medical Association shows that the country needs more than 50,000 critical care specialists. This implies that smart concepts need to be introduced to save the situation and fill the void.

“Problem of non-availability of quality care for critical patients in rural areas prompted us to design ICU Connect and NICU Connect solutions. During our innovation process, we collaborated with a world-leading neonatal equipment manufacturing company to start building solutions for intensive care units (ICUs), neonatal ICUs (NICUs) and neonatal care centres,” says Rajendra Kurady, co-founder and managing director, RTWO Healthcare Solutions.

Analytics in healthcare is at the forefront of providing a way of predicting life-threatening diseases. Millions of India's poor patients wait for days to gain admission to public hospitals. It is thus imaginable that potential for connected healthcare solutions, where people need not wait for their turns, are immense.

## Reinventing the invented

Innovation is a merger of ideas and thoughts to improve existing technology and making it way better than it is. Monitoring health vitals in real time may be at its infancy today, but it is about to go light years ahead as we speak. The innovation we are talking about here revolves around uberisation of medical device data.

The product connects the vital stats of humans that require intensive care to healthcare providers via servers, enabling monitoring of the patients in real time. Unlike traditional systems that merely focus on data acquisition, monitoring and reporting, this product can use predictive analytics to identify an issue proactively. It uses algorithms built into the device to give preventive alerts before the patient goes into crisis. Moreover, it has a small electronics board with an HL7 conversion kit to ease the documentation process.

“We have one among the highest mortality rates for any country in the world. About 70 per cent of infant deaths and more than half of under-five child deaths occur in the neonatal period. Manual data-capturing errors in ICUs and NICUs are not sufficient to enable accurate diagnosis and treatment due to the rapid changes in the condition of critical patients. This leads to deaths in critical care units (CCUs) or NICUs,” says Kurady.

At the heart of this innovation are crisis-prediction algorithms. These are built based on variation of some key vital health parameters like heart rate, temperature, respiratory rate and oxygen saturation to predict diseases like sepsis, RDS and asphyxia for neonates or adults. These parameters are captured from multipurpose bedside devices like baby warmers and ventilators. For example, when hypothermia or hyperthermia strikes a patient, there is a greater risk of getting sepsis as the patient panics while the heart rate elevates.

Trending indicators that monitor these heart rate variations indicate the progress while the system takes help of algorithms and smartly switches itself to the next level

**NICU Connect solution**

